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Small-Sample Bias in GMM Estimation of Covariance Structures

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We examine the small-sample properties of the generalized method of moments estimator applied to models of covariance structures, in which case it is commonly known as the optimal minimum distance (OMD) estimator. We find that OMD is almost always biased downward in absolute value. The bias arises because sampling errors in the second moments are correlated with sampling errors in the weighting matrix used by OMD. Furthermore, OMD is usually dominated by equally weighted minimum distance (EWMD). We also propose an alternative estimator that is unbiased and asymptotically equivalent to OMD. The Monte Carlo evidence indicates, however, that it is usually dominated by EWMD.

KEY WORDS: Generalized method of moments; IWOMD; Monte Carlo; Optimal minimum distance; Split sample.

Generalized method of moments (GMM) estimators have desirable asymptotic properties in many contexts, but little is known about their small-sample properties. In this article, we examine the small-sample properties of GMM for models of covariance structures. Recent applications of models of covariance structures include Abowd and Card's (1987, 1989) studies of contract models of employment and earnings growth, Hall and Mishkin's (1982) study of consumption, Behrman, Rozenzweig, and Taubman's (1994) study of the effects of individual endowments on earnings, own schooling, and spouse's schooling, and Griliches's (1979) survey of sibling models. In these contexts, GMM minimizes the weighted distance between sample moments and the implied population moments, in which case the weighting matrix is the inverse of a consistent estimate of the covariance matrix of the sample moments. In this context GMM is often referred to as the optimal minimum distance (OMD) estimator.

Our study of the small-sample properties of OMD is motivated by the fact that several authors report difficulties in empirical applications based on OMD estimation of covariance models, including Abowd and Card (1989) and Altonji, Martins, and Siow (1987). There have been similar reports of problems with GMM in empirical finance, where GMM has been widely adopted and researchers routinely work with models of second moments. For example, Lehmann (1990) reported difficulties in using feasible OMD to combine information from subsamples. Shanken (1990) estimated models of the mean and the variance of portfolio returns by ordinary least squares (OLS). He noted that simultaneous estimation of the models for the mean return and the variance offer advantages, but he reported in a footnote that "Attempts to incorporate the estimated residual variance relations in potentially more efficient WLS regression were unsuccessful and appeared to induce spurious associations that I do not fully understand" (p. 118).

To shed light on the difficulties with OMD encountered by empirical researchers, we present Monte Carlo evidence on the relative bias, variance, root mean squared error (RMSE), median absolute error (MAE), and coverage rates of the equally weighted minimum distance (EWMD) and OMD estimators for a set of covariance models. We argue that the correlation between sampling errors in the second moments and the sample weighting matrix generates bias in OMD. To isolate the weighting procedure as the sole source of bias, we focus on linear models of covariance structures, although many of the issues carry over to nonlinear models of population moments (Clark 1996). Our main finding is that OMD is seriously downward biased in absolute value in small samples for most distributions and in relatively large samples for poorly behaved distributions such as the lognormal. Furthermore, OMD usually has a larger RMSE and MAE than EWMD. The coverage rates of 90% confidence intervals for EWMD based on the usual asymptotic approximation are usually close to 90%, but those for OMD are often much less than 90%. We typically find that OMD outperforms EWMD in RMSE only in situations in which the RMSE of both estimators is small. EWMD typically dominates OMD even in designs in which the theoretical advantage of OMD is large. By comparing feasible OMD to OMD based on the theoretically optimal weighting matrix, we show that estimating the weighting matrix typically involves a large increase in sampling variance and in RMSE.

Our strongest evidence in favor of EWMD over OMD is based on the data used by Abowd and Card (1987, 1989). We begin with the "stationary model" of the growth in log earnings and log hours estimated by Abowd and Card. We show that OMD leads to substantial underestimates (in absolute value) of population second moments. Because the

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change in hours and earnings probably is not covariance stationary, we also present a simulation in which we treat the Abowd and Card data as the population and use the moments of that population to define a model that is true by construction. We analyze the distribution of the various estimators by drawing samples (with replacement) from the population and computing each estimator on the sample. The results strongly reinforce our main theme, which is that the OMD estimator of covariance structures suffers from serious downward bias in absolute value and that OMD is usually dominated by EWMD. The best choice for the Abowd and Card data is EWMD, which is what they chose on pragmatic grounds.

In addition to comparing EWMD and OMD, we propose and investigate an alternative estimator, which we call independently weighted optimal minimum distance (IWOMD). IWOMD is a split-sample estimator that uses separate groups of observations to estimate the moments and the weights. The random partitioning of the data breaks the sampling covariance between the moments and the weights. Parameter estimates are computed separately for each partition, and the estimates are averaged to form a final parameter estimate. IWOMD has identical large-sample properties to the OMD estimator but is unbiased regardless of sample size. The Monte Carlo evidence indicates, however, that IWOMD is usually dominated by EWMD based on the criteria of RMSE, MAE, and confidence-interval coverage rates.

The article is organized as follows. In Section 1 we illustrate the class of models of interest with an example that underlies some of our simulations. We define the OMD and EWMD estimators for the problem and present initial Monte Carlo evidence indicating that OMD is downward biased. In Section 2 we provide a theoretical discussion of the bias in OMD. The discussion points to an inverse relationship between the size of the bias and the precision in the second moments, which is strongly confirmed by the Monte Carlo evidence. In Section 3 we present the IWOMD estimator. In Section 4 we present more detailed Monte Carlo evidence on the performance of EWMD, OMD, and IWOMD. In Section 5 we present the empirical example based on the data used by Abowd and Card (1987, 1989) and also briefly discuss evidence from Schwert and Seguin (1990). We close the article with a brief summary and a research agenda. We wish to stress that care is required in extrapolating our results beyond the models of covariance structures we study. Although our points that the correlation between the moments being fitted and the weights will lead to bias and that imprecision in the weights will reduce or even eliminate theoretical efficiency gains are general, GMM may perform well in time series applications involving first moments, cases in which it has frequently been used.

1. EVIDENCE OF SMALL-SAMPLE BIAS IN OPTIMAL MINIMUM DISTANCE

In the basic covariance model, multiple sample moments are combined into a single estimate of the population mo-

ments. Our examples focus on linear models relating sample variances (or covariances) to a single population parameter so that we isolate a particular source of small-sample bias. Assume that the researcher has data observations D_{pi} , where $p = 1, \dots, P$ indexes the variable used to compute the sample moment and $i = 1, \dots, N_p$ indexes the observations on that variable. For each value of p , the mean and variance are computed using the standard unbiased estimators

$$\bar{D}_p = \frac{1}{N_p} \sum_{i=1}^{N_p} D_{pi} \quad (1)$$

and

$$m_p = \frac{1}{(N_p - 1)} \sum_{i=1}^{N_p} (D_{pi} - \bar{D}_p)^2; \quad E(m_p) = \mu_p. \quad (2)$$

The second moment estimates are stacked into a $(P \times 1)$ vector, m , and are related to a $(P \times 1)$ vector of population moments, μ , through the model

$$m = \mu + \varepsilon = f(\theta) + \varepsilon. \quad (3)$$

In (3), θ is the $(Q \times 1)$ parameter vector one wishes to estimate and ε is a $(P \times 1)$ vector of sampling errors. When $f(\theta)$ is linear in θ , the model is

$$m = X\theta + \varepsilon. \quad (4)$$

For example, suppose that a researcher wishes to estimate a population variance from observations on a panel of individuals covering 10 time periods. In this case, $P = 10$, and the vector m contains 10 estimates of the variance, one from each time period. The matrix X is a (10×1) vector of ones, and θ is the population variance (a scalar). EWMD amounts to a least squares regression of m on X with the familiar solution

$$\begin{aligned} \theta_{\text{EWMD}} &= \text{ArgMin}_{\theta} (m - f(\theta))'(m - f(\theta)) \\ &= (X'X)^{-1}(X'm). \end{aligned} \quad (5)$$

The EWMD estimator is not efficient if the elements of ε are heteroscedastic or correlated. Heteroscedasticity may arise as a result of unbalanced data (N_p may differ across p) or because the distributions of the D_{pi} are different. The elements of ε will be correlated if $\text{cov}(D_{pi}, D_{p'i})$ is not 0, which is likely in panel-data applications. The variance of ε is not a scalar matrix in these cases. OMD takes this into account and is generalized least squares (GLS) applied to (4). The covariance matrix of ε , Ω , is replaced with a consistent estimate obtained from the same data used to compute the sample second moments m . For example, a conventional estimator of the variance of m_p is

Table 1. Performance of Estimators of a Variance Parameter Using 10 Sample Variances From 1 Distribution

		Equally weighted (EWMD)				Optimally weighted (OMD)					Independently weighted (IWOMD)				
		Std	90% coverage			90% asym.					Std	90% coverage			
Row distribution	Obs.	RMSE	MAE	Asym.	Boot	Bias	Std	RMSE	MAE	coverage	Bias	RMSE	MAE	Asym.	Boot
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1 <i>t</i> (5)	50	.126	.071	.84	.79	−.199	.101	.223	.201	.15	−.011	.202	.125	.41	.75
2 <i>t</i> (10)	50	.079	.052	.89	.86	−.118	.089	.148	.121	.41	.000	.136	.080	.57	.82
3 <i>t</i> (15)	50	.073	.050	.90	.86	−.100	.083	.130	.101	.47	−.001	.116	.077	.60	.83
4 Normal	50	.063	.041	.90	.86	−.074	.075	.105	.079	.57	−.003	.099	.065	.62	.81
5 Uniform	50	.042	.029	.90	.86	−.013	.048	.049	.032	.80	−.002	.055	.036	.75	.90
6 Lognormal	50	.395	.206	.74	.71	−.616	.128	.629	.625	.00	−.024	.899	.377	.12	.76
7 Exp	50	.125	.085	.88	.86	−.279	.131	.309	.281	.13	−.012	.255	.165	.31	.85
8 Half-normal	50	.074	.050	.88	.85	−.118	.088	.147	.120	.41	−.001	.131	.084	.52	.84
9 Bimodal	50	.039	.027	.90	.84	−.021	.044	.049	.034	.75	−.001	.053	.035	.73	.88
10 <i>t</i> (5)	100	.083	.057	.88	.85	−.124	.070	.143	.126	.28	.003	.126	.077	.57	.70
11 <i>t</i> (10)	100	.054	.038	.89	.87	−.065	.059	.087	.067	.55	.000	.073	.050	.69	.78
12 <i>t</i> (15)	100	.051	.035	.88	.86	−.052	.055	.076	.056	.62	−.001	.068	.046	.72	.78
13 Normal	100	.043	.029	.90	.88	−.034	.047	.058	.039	.70	.003	.055	.037	.75	.79
14 Uniform	100	.029	.020	.89	.86	−.006	.031	.031	.021	.85	−.001	.033	.021	.83	.88
15 Lognormal	100	.344	.156	.78	.77	−.474	.115	.488	.480	.01	.016	.752	.254	.19	.73
16 Exp	100	.085	.055	.86	.85	−.165	.093	.189	.169	.27	−.003	.147	.095	.50	.75
17 Half-normal	100	.052	.034	.90	.89	−.062	.059	.085	.066	.57	−.001	.074	.051	.68	.78
18 Bimodal	100	.027	.018	.87	.84	−.011	.029	.031	.020	.77	−.001	.031	.021	.77	.80
19 <i>t</i> (5)	500	.042	.025	.88	.88	−.041	.034	.054	.043	.57	.002	.067	.031	.70	.78
20 <i>t</i> (10)	500	.025	.017	.89	.88	−.014	.025	.029	.021	.79	.001	.027	.018	.82	.84
21 <i>t</i> (15)	500	.023	.016	.90	.89	−.011	.024	.026	.018	.82	.000	.025	.018	.84	.84
22 Normal	500	.019	.012	.90	.90	−.007	.020	.021	.013	.87	.000	.020	.013	.88	.88
23 Uniform	500	.012	.008	.89	.88	−.001	.012	.012	.008	.89	.000	.012	.008	.89	.88
24 Lognormal	500	.147	.080	.82	.82	−.233	.084	.247	.236	.08	.008	.238	.118	.40	.68
25 Exp	500	.041	.025	.91	.90	−.043	.043	.061	.047	.64	−.001	.051	.033	.76	.81
26 Half-normal	500	.024	.017	.90	.90	−.012	.024	.027	.020	.83	.002	.027	.019	.86	.87
27 Bimodal	500	.012	.008	.90	.90	−.002	.012	.012	.008	.90	−.000	.012	.008	.91	.90
28 <i>t</i> (5)	1,000	.029	.018	.89	.87	−.026	.025	.036	.027	.624	−.001	.033	.021	.74	.78
29 <i>t</i> (10)	1,000	.018	.012	.90	.89	−.008	.018	.020	.013	.829	−.000	.019	.013	.87	.87
30 <i>t</i> (15)	1,000	.016	.011	.89	.88	−.006	.016	.017	.012	.840	−.001	.017	.011	.87	.87
31 Normal	1,000	.014	.009	.90	.89	−.004	.014	.015	.010	.887	−.000	.014	.010	.89	.86
32 Uniform	1,000	.009	.006	.92	.91	−.001	.009	.009	.006	.896	−.000	.009	.006	.91	.91
33 Lognormal	1,000	.100	.060	.84	.84	−.161	.067	.175	.165	.138	.002	.151	.085	.45	.70
34 Exp	1,000	.030	.020	.89	.88	−.020	.031	.037	.026	.752	.003	.035	.023	.82	.84
35 Half-normal	1,000	.017	.011	.91	.90	−.006	.017	.018	.013	.861	.001	.018	.012	.88	.88
36 Bimodal	1,000	.008	.006	.90	.90	−.000	.008	.008	.006	.904	.001	.008	.006	.89	.89

NOTE: 1. Results are based on 1,000 replications of a model that equates 10 uncorrelated variances from the specified distribution and sample size to a single population variance. 2. The columns labeled "90% coverage" report the probability that the symmetric 90% confidence-interval estimate contains (covers) the true value of 1. "Asym." indicates that the confidence-interval estimate is based on the conventional asymptotic formula. "Boot" indicates that a bootstrap confidence interval is used. We constructed the bootstrap confidence interval for EWMD as follows. For each of 1,000 Monte Carlo samples of size N , we draw with replacement 500 different bootstrap samples of size N . We then compute the EWMD estimator using each of the 500 bootstrap samples. The 5th and 95th values of the EWMD estimator form the 90% confidence-interval estimate for the particular Monte Carlo sample. We use an analogous bootstrap procedure to produce column (14) for IWOMD.

$$\begin{aligned} \text{var}(m_p) &= \frac{N_p^2}{(N_p - 1)(N_p - 2)^2} \\ &\times \left[\frac{1}{N_p} \sum_{i=1}^{N_p} (D_{pi} - \bar{D}_p)^4 \right. \\ &\quad \left. - \left(\frac{1}{N_p} \sum_{i=1}^{N_p} (D_{pi} - \bar{D}_p)^2 \right)^2 \right]. \quad (6) \end{aligned}$$

A comparable formula exists for the covariance between m_p and $m_{p'}$. Let $\hat{\Omega}$ represent the estimated covariance matrix of m . The OMD estimator minimizes a quadratic form involving the sampling error in the moments and $\hat{\Omega}$. In other

words,

$$\begin{aligned} \theta_{\text{OMD}} &= \text{ArgMin}_{\theta} (m - f(\theta))' \hat{\Omega}^{-1} (m - f(\theta)) \\ &= (X' \hat{\Omega}^{-1} X)^{-1} X' \hat{\Omega}^{-1} m \end{aligned} \quad (7)$$

in the linear case.

Malinvaud (1970), Chamberlain (1982, 1984), Hansen (1982), and others showed that under a relatively innocuous set of conditions $N^{1/2}(\theta_{\text{OMD}} - \theta) \rightarrow_D N(F' \Omega^{-1} F/N)$, where F is $\partial f / \partial \theta$ evaluated at the true θ , and that the OMD estimator is asymptotically efficient.

The replacement of Ω with a consistent estimate does not affect the asymptotic properties of θ_{OMD} . It does, however, affect the small-sample behavior of the estimator. Table 1 examines the EWMD and OMD estimators of the 10-variance model mentioned previously, using simulated data from several common distributions and several sample

sizes. We consider the case in which the data D_{pi} are independent across i and p , which implies that the 10 sample variances are independent. The distributions were chosen to include thick-tailed symmetric distributions [student- $t(5)$ and student- $t(10)$], long-tailed asymmetric distributions (exponential and lognormal), an asymmetric distribution (half-normal), several "well-behaved" distributions (normal and uniform), and a bimodal distribution. (The bimodal distribution is generated by using a binomial to choose randomly between two normally distributed random variables. One has mean -2 , one has mean 2 , and both have variance 1 .) All of the population distributions are scaled to have mean 0 and variance 1 . For each distribution, a fixed number of observations is drawn at random and used to estimate the 10 variances and their covariance matrix Ω . Both the diagonal and off-diagonal elements of Ω are replaced with sample estimates, even though the elements of m are independent. This corresponds to the case in which the econometrician is unaware of the fact that the moments are independent. The EWMD and OMD estimates are computed and the entire process is replicated 1,000 times.

The columns in the first panel of the table (rows 1–9) that are labeled "Bias" report the average of $(\hat{\theta} - \theta)$ from 1,000 replications of the experiment based on 50 observations per moment ($N_p = 50, p = 1, \dots, 10$). We omit the column for EWMD because it is known to be unbiased. Column (5) shows that OMD has a negative bias. For example, when data are drawn from the standard normal distribution, the OMD estimates have a bias of $-.074$, or 7.4% of the true parameter value of 1. The mean of OMD when using data from the uniform distribution is still .013 below the true value.

The bias in OMD is larger for thick-tailed distributions, which have higher fourth moments and therefore have more variable second moments. To see this, note that the bias declines with the number of degrees of freedom of the t distribution, and recall that the tails of the student- t distribution become thinner as the degrees of freedom increase and the distribution approaches the normal. For example, when N is 50 the bias in the OMD estimator declines from $-.199$ to $-.118$ to $-.100$ as one moves from a $t(5)$ distribution to a $t(10)$ and then to a $t(15)$ distribution. The lognormal and exponential distributions produce the worst bias. The bias is $-.616$ and $-.279$, respectively, when N is 50, which is very large relative to the true parameter value of 1.

The remaining panels of the table repeat the experiment, increasing the amount of data used to estimate each moment. As N rises, EWMD remains unbiased and the bias in OMD declines. The additional data improve the accuracy of the variance estimates and the accuracy of the weights. This, in turn, improves the accuracy of the OMD estimator. For most distributions the bias is very small when 1,000 observations are available to estimate each sample moment. The bias is 16% in the lognormal case, however, even with 1,000 observations per moment.

Table 1 suggests three conclusions. First, OMD is downward biased. Second, the bias dissipates with sample size.

Third, the bias is worse in long-tailed and skewed distributions.

Because there is no heteroscedasticity or serial correlation in the experiments of Table 1, the EWMD estimator is the optimally weighted estimator, and OMD and EWMD are asymptotically equivalent. Thus, it is not surprising that EWMD outperforms OMD in these cases. The detailed tables of Altonji and Segal (1994) (hereafter AS) show, however, that the three conclusions about bias hold when heteroscedasticity is introduced by varying the distribution of the data within an experiment and when OMD is used to estimate a parameter of the covariance matrix of a vector of correlated random variables. Section 3 provides a graphical summary of some of this evidence.

2. THE SOURCE OF BIAS IN OMD

Feasible GLS estimation, including OMD, is justified by its asymptotic properties rather than small-sample properties. Bias will arise in most cases in which the estimate of the weighting matrix covaries with the regression model error.

This problem can be serious in the case of OMD estimation of covariance structure models. From (7) the bias for fixed N is

$$E(\theta_{\text{OMD}} - \theta) = E[(X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}\varepsilon]. \quad (8)$$

In OMD the sample estimate, $\hat{\Omega}$, of the covariance matrix of the vector of sample moments, m , is the weighting matrix. When m consists of second moments, it is likely to be correlated with $\hat{\Omega}$ because individual observations that increase the sample estimate of a variance will also tend to increase the sample estimate of the variance of the variance. The resulting correlation between the elements of $\hat{\Omega}$ and ε creates a correlation between the elements of $(X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}$ and the elements of ε . Thus, the bias is nonzero even though the mean of the error ε in the sample moments m is 0.

The consequences of correlation between the weights and the errors is easily illustrated in a model of independent but heteroscedastic sample moments. Let ω_p be the estimated variance of the p th element of m , m_p , and assume that $E(m_p) = \theta$ for $p = 1, \dots, P$. The design matrix for this model is a vector of ones, so the bias of the OMD estimator is

$$E(\theta_{\text{OMD}} - \theta) = E \left[\left(\sum_{p=1}^P \omega_p^{-1} \right)^{-1} \sum_{p=1}^P \omega_p^{-1} \varepsilon_p \right]. \quad (9)$$

The sign of the bias in θ_{OMD} for a fixed sample size N requires an examination of the expectation in (9). The first summation is the sum of inverses of the variance estimates and must be positive. Thus, the sign of the bias depends on the weighted sum of the ε_p in the second summation. Intuitively, the weight ω_p^{-1} (proportional to the inverse of the difference between the fourth moment and the square of the second sample moment) tends to be small when there are unusually large or small observations in the sample used to compute the sample variance m_p , but ε_p (the sampling error in m_p) tends to be positive. Consequently, the estimator

gives less weight to positive realizations of the sampling error ε_p than to negative realizations. Because ε_p has a mean of 0 and the estimator places less weight on positive values of ε_p , the bias is negative.

Formally evaluating the expectation on the right side of (9) is difficult because of the correlation between the terms of the first and second summations. Several studies (e.g., Angrist and Krueger 1995; Bekker 1994), however, have evaluated small-sample properties in other contexts by allowing the number of moments to grow with the sample size. Using a similar approach we evaluate the probability limit of θ_{OMD} as the number P of independent sample variances of *fixed sample size* N goes to infinity. Dividing each summation by P , the probability limit of θ_{OMD} as P goes to infinity is

$$\text{plim}_{P \rightarrow \infty} (\theta_{\text{OMD}} - \theta) = E(\omega_p^{-1})^{-1} E(\omega_p^{-1} \varepsilon_p), \quad (10)$$

provided the expectations exist.

The random variable ω_p is greater than 0 with probability 1, so the first expectation in (10) is positive. Thus, the sign of the bias is determined by the sign of the second expectation. Because ω_p is greater than 0 with probability 1, ω_p^{-1} is strictly decreasing in ω_p . This suggests that $E(\varepsilon_p/\omega_p)$ will be opposite in sign from $E(\varepsilon_p \omega_p)$, although the monotonicity of the transformation ω_p^{-1} is not sufficient to guarantee this. That is, if $\text{cov}(\varepsilon_p, \omega_p) > 0$, then $E(\varepsilon_p/\omega_p)$ is likely to be less than 0 and the bias in OMD is negative. In the Appendix we show that $\text{cov}(\varepsilon_p, \omega_p)$ is greater than 0 if the distribution of $(D_{ip}^2 - E(D_{ip}^2))$ has a positive skew. This condition holds for the standard distributions used in the economics literature, the distribution of the Abowd-Card data, and for all of the distributions used in Table 1. Thus, the analysis of (10) supports the intuitive argument for a negative bias and the Monte Carlo evidence of Table 1. We provide an example in the Appendix in which $\text{cov}(\varepsilon_p, \omega_p) < 0$ and the bias in OMD is positive. The example involves a symmetric distribution with finite support and most of the mass near the minimum and maximum values.

It is interesting to relate this result to feasible GLS applied to problems involving means or conditional means. If the moments $m_p, p = 1, \dots, P$, were the sample means rather than the variances of D_{ip} and θ was the population mean, then the exact same analysis implies a negative (positive) bias if D_{ip} has a positive (negative) skew. The OMD weights for combining the means of D_{ip} from the P groups into an estimate of the population mean θ are the sample variances for each group p . Thus the weights are even functions of the vector $\{D_{1p}, \dots, D_{Np}\}$. If D_{ip} are independent and have symmetric distributions then the sampling error in the sample mean m_p will have a symmetric distribution. This symmetry condition together with the fact that the OMD weights are even functions are basically the Kakwani (1967) conditions for feasible GLS regression to be unbiased. In the preceding problem involving a model of variances (rather than means) the weight ω_p is also an even

function of $\{D_{1p}, \dots, D_{Np}\}$, but ε_p has a symmetric distribution only if $(D_{ip}^2 - E(D_{ip}^2))$ has a symmetric distribution.

Our simulation results using the Abowd and Card data and other unreported Monte Carlo experiments indicate that OMD estimates of covariances, like variances, are often biased downward in absolute value. To get some insight into this, we have formulated a comparable model with m_p redefined to be the covariance between two mean 0 random variables Z_{ip} and Y_{ip} . The sign of the bias continues to depend on $E(\varepsilon_p/\omega_p)$, where ε_p and ω_p are defined from the model of covariances. $E(\varepsilon_p/\omega_p)$ is likely to be of opposite sign to $\text{cov}(\varepsilon_p, \omega_p)$. One may show that $\text{cov}(\varepsilon_p, \omega_p)$ has the same sign as $E[(YZ - E(YZ))^3]$. Because the cross-products of positively (negatively) correlated variables are likely to have a positive (negative) skew, the simple theoretical analysis is consistent with the simulation results.

What determines the size of the bias? The intuitive argument implies that it will depend on the degree to which ε_p covaries with ω_p , which suggests that the bias is lower for large N and higher for distributions with larger higher-order moments because these affect $\text{var}(\varepsilon_p)$. To see the dependence on N , note that standard asymptotic analysis of OMD examines the behavior of the estimator for fixed P as N goes to infinity and therefore implies that the bias in θ_{OMD} declines with the sample size per moment. The distributions of ω_p and m_p are implicitly indexed by N and the distribution of D_{pi} . As N goes to infinity, ε_p converges in probability to 0 and the covariance between ε_p and ω_p^{-1} dissipates. Therefore $E(\varepsilon_p/\omega_p)$ normalized by $E(1/\omega_p)^{-1}$ converges to 0. For a given N , $\text{var}(\varepsilon_p)$ and $\text{cov}(\varepsilon_p, \omega_p)$ are positive functions of the higher moments of the distribution, leading to larger bias for distributions with larger higher-order moments. The bias may be small in situations in which the sampling errors ε_p are small, either because N is large or because the underlying distributions are well behaved. In these situations, however, any improvements in precision from using OMD are also likely to be small in absolute magnitude.

In Figure 1, we examine the implication that there is a relationship between $\text{var}(\varepsilon_p)$ and the bias, $E(\theta_{\text{OMD}} - \theta)$, by plotting the theoretical standard deviation (std) of the sample variance against the size of the downward bias of θ_{OMD} for the Monte Carlo experiments of Table 1. The theoretical standard deviation is the square root of the variance of the variances and is a function of the sample size and the population moments of the particular distribution. The bimodal and half-normal distributions are omitted from the figure because we have not calculated the theoretical variance of the variance in these cases. The figure shows a strong positive relationship between the downward bias in θ_{OMD} and the std of the sample variances. The solid regression line in the figure has a slope of .46 and an adjusted R -square of .95. As the sample size declines and as the fourth moment of the underlying distribution increases holding the second moment fixed, the variance of the variance increases and the bias becomes worse. The tight fit in the figure suggests that the simple relationship between the bias and the theoretical std holds regardless of whether the variation in the theoretical std is due to variation in the distribution across

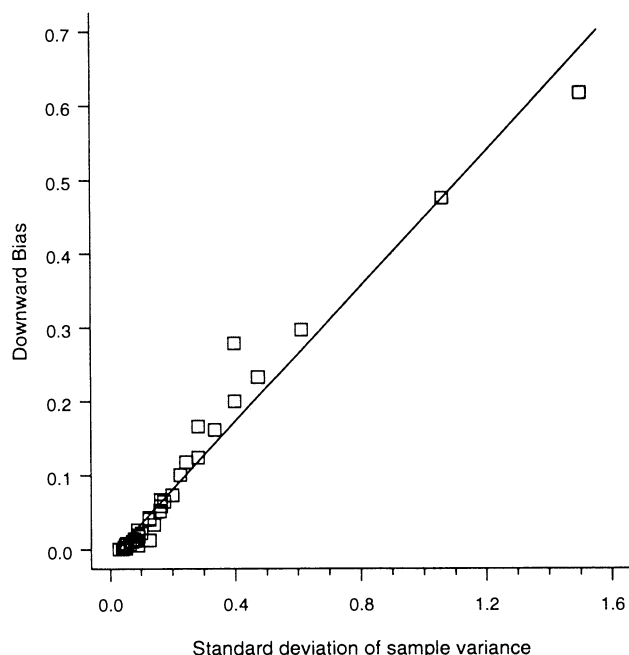


Figure 1. Relationship Between Sampling Variance and OMD Bias: Based on 10 Moments From 7 Distributions and 5 Sample Sizes; Predicted Bias = $-.01 + .46 \times \text{Std of Sample Variance}$; Adjusted R-Squared is .95.

experiments or due to variation in sample size across experiments.

The preceding results suggest that research on the small-sample properties of GMM and alternative estimators is needed. Before proceeding further, it is worth mentioning a few recent works that have considered the small-sample properties of GMM. Koenker, Machado, Skeels, and Welsh (1994) provided a theoretical analysis of small-sample properties of GMM in a somewhat different context than ours but focused on efficiency rather than bias. They suggested alternatives based on robust estimators rather than the least squares criteria that underly conventional GMM. Arellano and Sargan (1990) discussed alternative small-sample approximations to the distribution of estimators that may be written as functions of second moments, with references to earlier works, but did not apply the approximations to the class of problems we consider. Tauchen (1986) examined the use of GMM to estimate parameters of nonlinear models of conditional means with endogenous variables in time series data. Horowitz and Neumann (1992) presented a second-order correction for bias in a GMM-based test statistic of the proportional-hazards assumption in duration analysis. Neither these studies nor a handful of other recent working papers referenced by Ogaki (in press) raised the issue of bias from correlation between the moments under study and the weighting matrix, which is our focus.

3. AN UNBIASED OMD ESTIMATOR

The bias in OMD arises because of a correlation between sample moments and the estimated weighting matrix $\hat{\Omega}$. Corrections based on bias approximations are cumbersome. We develop an unbiased estimator based on the old idea in econometrics of splitting the sample to break statis-

tical dependence. [Another recent application of this idea is the article by Angrist and Krueger (1995).] Observe that estimates of the second moments based on part of the sample are statistically independent of estimates of the weighting matrix based on the other part. Consider a random partitioning of the full data sample into G groups of equal size. Let the $(P \times 1)$ vector m_g and the $(P \times P)$ matrix $\hat{\Omega}_g$ be the sample estimates of μ [recall that $\mu = f(\theta)$] and Ω using only the data in group g . Let $m_{(g)}$ and $\hat{\Omega}_{(g)}$ represent estimates based on the data excluding group g . Define the IWOMD estimator $\theta_{\text{IWOMD}(G)}$ as the split-sample estimator that uses each m_g and $\hat{\Omega}_{(g)}$ pair to produce G separate parameter estimates, which are then averaged. The formal definition of the IWOMD estimator is

$$\theta_{\text{IWOMD}(G)} = \frac{1}{G} \sum_{g=1}^G \text{ArgMin}_{\theta} (m_g - f(\theta))' \hat{\Omega}_{(g)}^{-1} (m_g - f(\theta)). \quad (11)$$

For $f(\theta)$ linear in θ , each term in the average is free of small-sample bias because the sampling errors in the estimates of μ and Ω are independent. Hence the average is unbiased. In the linear case, θ_{OMD} and $\theta_{\text{IWOMD}(G)}$ are asymptotically equivalent estimators. This is easily demonstrated using (11). The basic argument is that the $\hat{\Omega}_{(g)}$ and $\hat{\Omega}$ used in OMD all converge to Ω . When $\hat{\Omega}_{(g)}$ is set to Ω for all g , θ_{IWOMD} is numerically identical to OMD based on the true Ω . Consequently, θ_{IWOMD} has the same limiting distribution as feasible OMD and true OMD.

Note that the estimates m_g and $\hat{\Omega}_{(g)}$ will be dependent if a single sample mean (or regression function) is used to center the data, as is often done in EWMD and OMD. Consequently, IWOMD is not unbiased in small samples unless the data are centered separately for partitions g and (g) with an appropriate degree-of-freedom adjustment in computing m_g and $\hat{\Omega}_{(g)}$. When we refer to IWOMD as unbiased, we are referring to the estimator in which either the mean is known or the data are centered separately. Estimation of the mean has no asymptotic effect, so researchers often ignore the effects of estimating the means in actual applications. Our simulations (not reported) indicate, however, that the bias due to correlation through the mean may be substantial if the sample size is small and the distribution is heavily skewed like the lognormal and the exponential distributions. Use of a single mean to center the data produces positive bias in the IWOMD estimator.

We investigated whether the number of partitions G matters much. The number of groups affects the estimation by trading off the precision of the moment estimates and the weighting estimates. The precision loss in the moment estimates due to increased G is offset by the averaging of the parameter estimates as the last step in the procedure. When $f(\theta)$ is nonlinear in θ , however, the average of the estimates of θ based on each of the subgroups of observations may be biased as a consequence of Jensen's inequality. One could use IWOMD to estimate the most restrictive linear model that nests the nonlinear model and then use EWMD to fit

the nonlinear model to the parameter estimates of the linear model.

In Monte Carlo simulations (not reported) we compared the performance of the estimator when G is set to 2, 5, 10, 25, and N groups when the data are not centered separately for each partition. The simulations suggest that its performance usually is not very sensitive to choice of G . We use $G = 2$ throughout the article, however, out of a concern that in actual applications centering the data separately for each partition may cause problems when the ratio of N to G is small.

In Table 1 we report simulations of θ_{IWOMD} analogous to those of θ_{EWMD} and θ_{OMD} discussed previously. The results in column (10) confirm that θ_{IWOMD} is unbiased.

4. EVALUATING THE PERFORMANCE OF EWMD, OMD, AND IWOMD

In this section we provide a broader assessment of the performance of EWMD and OMD. We consider the bias, RMSE, MAE, and std of the estimators. We also consider the coverage rates of 90% confidence-interval estimates. We defer discussion of IWOMD until the end of the section.

4.1 Models With Identically Distributed Moments

In Table 1 we report the std, RMSE, and MAE of EWMD and OMD estimators of the variance parameter θ as well as the bias. In these experiments EWMD is OMD based on the true weights. The table shows that OMD is not only biased but always has an RMSE and an MAE that are as large or larger than EWMD. For example, in the case of the $t(5)$ with 50 observations, the bias in OMD is $-.199$. It has an RMSE of $.223$, almost double that of EWMD.

Given that EWMD is “true” OMD in these experiments, it is initially surprising that, for the $t(5)$ and the lognormal cases, OMD has a *smaller* std than EWMD. The source of bias in OMD—the correlation between the weight and the second moment—is expected to reduce the sampling variance of OMD when the kurtosis of the underlying data is large (Koenker et al. 1994), as in the $t(5)$ and lognormal cases. These are typically the cases in which bias is large.

The fact that sampling error in $\hat{\Omega}$ leads to bias in OMD raises concern about the use of the conventional asymptotic t statistics, standard errors, and confidence intervals as the basis of statistical inference in the EWMD or IWOMD cases because these rely heavily on $\hat{\Omega}$. Moreover, to the extent that departures from normality in the second moments and the estimators are substantial, confidence intervals and test statistics based on the asymptotic standard errors must be treated cautiously.

To assess these concerns, we report the fraction of the replications in which the 90% confidence-interval estimate for θ covers the true value of 1 in columns (3) and (9) of Table 1. The confidence-interval estimates were constructed using $\hat{\Omega}$ to form the conventional asymptotic standard errors for both EWMD and OMD. The coverage rate is close to .9 or slightly below in the case of EWMD for sample sizes of 50, 100, and 500, except when the lognormal is used in the experiment. In contrast, the coverage rates for the OMD

confidence intervals are well below .9 in most cases, even when N is 500. For example, when N is 500 the OMD coverage rates for the $t(5)$, exponential, and lognormal are .57, .64, and .08.

Given concerns about using the estimated weighting matrix to form confidence intervals, we also examined the performance of a 90% symmetric bootstrap confidence interval for the EWMD simulations [col. (4)]. These results indicate that the use of $\hat{\Omega}$ with the asymptotic standard-error formula is usually a bit better than inference based on the bootstrap. For example, for the $t(5)$ with 100 observations, the coverage rate using the conventional asymptotic standard errors is .88, and coverage rate of the bootstrap confidence interval is .85. The coverage rates of bootstrap confidence intervals for OMD are usually even worse than those based on the asymptotic theory, and we do not report them.

In summary, we find first that inference based on the use of $\hat{\Omega}$ is satisfactory in most cases when the estimator is EWMD. Difficulties arise in the cases involving the lognormal. Second, we find that the coverage rates for the OMD 90% confidence intervals are often far below the nominal value of .9.

4.2 Fitting Variance Parameters to Second Moments Based on Different Distributions

AS reported Monte Carlo experiments when 5 moments come from one distribution and 5 from another for most combinations of the distributions in Table 1 and sample sizes of 50, 100, 300, 500, and 1,000. Figure 2, a–d, summarizes the evidence on the relative performance of the estimators. For the simulation experiments with similar theoretical efficiency gains (between 1.125 and 1.5), we plot the bias squared (squares and solid regression line) and the difference between the variance of EWMD and OMD (circles and dashed regression line) against the average variance of the sample variance. (We will discuss the dotted line later.) The pairs of distributions used are the $t5$ – $t10$, $t5$ – $t15$, $t15$ –uniform, uniform–normal, $t10$ –exponential, and $t15$ –exponential. The mean-squared errors of the EWMD and OMD are the same where the solid and dashed lines intersect. For a fixed efficiency gain, the OMD bias increases much faster than $\text{var}(\theta_{OMD}) - \text{var}(\theta_{EWMD})$ as the variance of the sample variances increases. The figure suggests that the realized efficiency gain of OMD dominates the bias only for very well-behaved distributions with small fourth moments. In Figure 2b we exclude experiments with high values for the average variance of the sample variance to be sure that these are not dominating the figure. Figure 2c and Figure 2d are analogous to 2a and 2b (respectively) but consider cases in which the theoretical efficiency gain is between 3.03 and 4.04. [The pairs of distributions are the uniform– $t(5)$, the uniform–exponential, the $t(5)$ –lognormal, and the exponential–lognormal.] In these figures the dashed line is positively sloped, indicating that $\text{var}(\theta_{EWMD}) - \text{var}(\theta_{OMD})$ is positively related to the average variance of the sample variances. The bias squared line (the solid line) has a much larger slope, however. The bias squared usually lies above $\text{var}(\theta_{EWMD}) - \text{var}(\theta_{OMD})$,

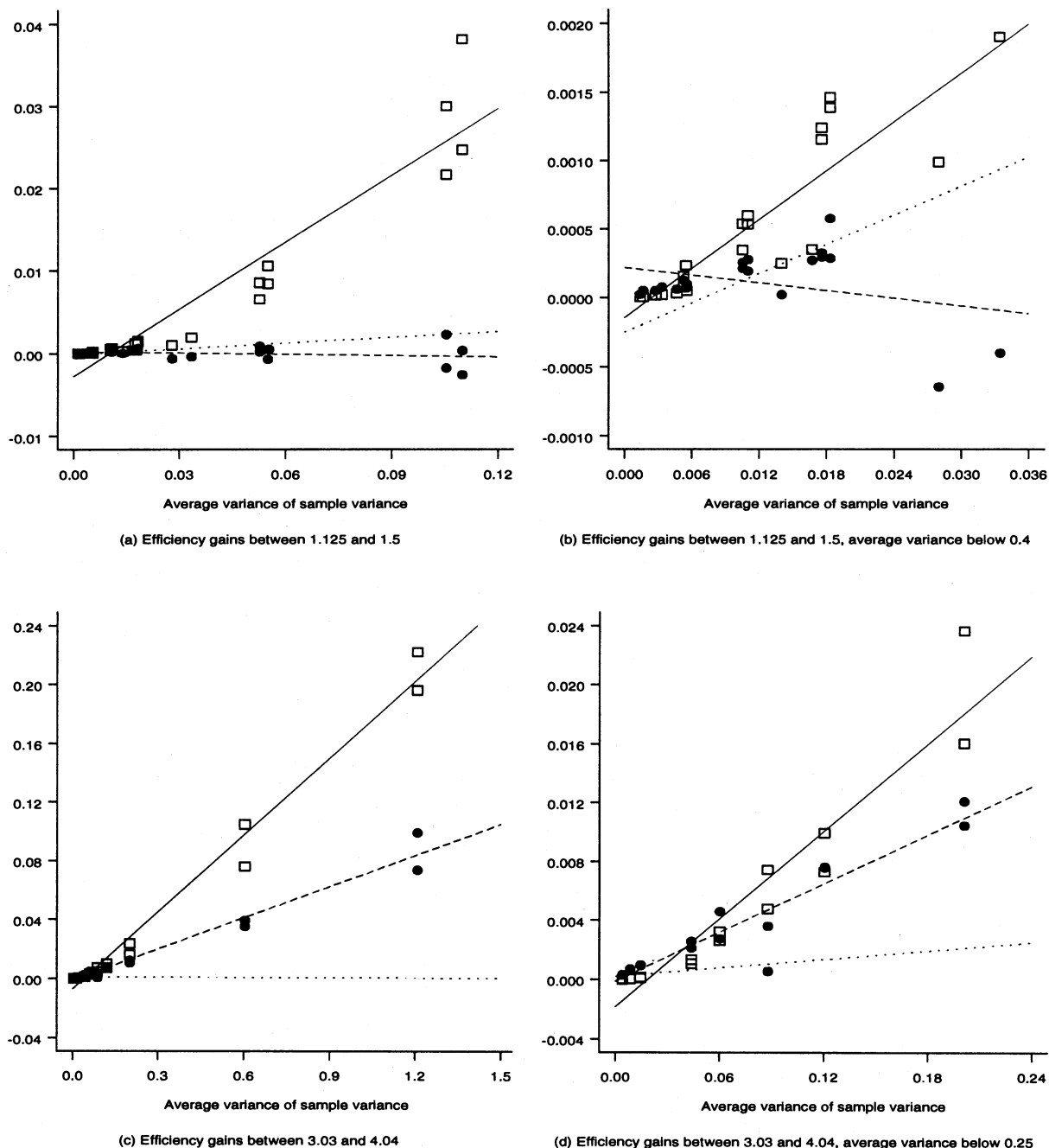


Figure 2. Decomposition of the Mean Squared Error of EWMD and OMD. Squares and solid lines indicate bias squared. Circles and dashed lines indicate the difference between the variance of EWMD and the variance of OMD. The dotted lines indicate the difference between the variance of feasible OMD and the variance of the true OMD.

except in cases in which average variance of the sampling variances is small and efficiency makes little difference in absolute terms.

The evidence in the figures suggests that the advantage of OMD in RMSE over EWMD (when there is an advantage) is usually only a small fraction of the theoretical advantage. To get further insight, we also compare the performance of feasible OMD to the performance of "true" OMD based on a known, optimal weighting matrix. The dotted line in the figures is the least squares regression line relating the difference between the sampling variances of OMD and "true OMD" based on a known, optimal weighting matrix to the

average variance of the sample variances. The sum of the dotted line and the solid line is the difference in RMSE between OMD and true OMD. The gap in RMSE between OMD and true OMD arises because of bias in OMD and because in most cases OMD has a higher sampling variance. The gap in sampling variance is reflected in the figures, where the dotted line graphing $\text{var}(\theta_{\text{OMD}})$ minus the variance of true OMD lies above 0. Both bias and sampling variance are more important when the theoretical efficiency gain from OMD is high (e.g., the lognormal-normal and lognormal-uniform cases). Bias is much more important when the theoretical efficiency gain is low but the distributions are badly behaved [e.g., the exponential- $t(5)$]. The

high cost of estimating the weighting matrix suggests that there are large gains from a priori information about the relative precision of different sample moments.

AS also examined coverage rates for some of the experiments involving two distributions. EWMD continues to dominate OMD in RMSE and MAE and in the coverage rates of confidence intervals. The coverage rates of EWMD confidence intervals based on the asymptotic theory are close to .9, except in cases involving the lognormal. OMD coverage rates are often well below .9.

4.3 Experiments Involving Correlated Moments

AS also presented detailed Monte Carlo evidence on the behavior of the estimators when the sample moments are correlated. To save space, we simply describe the experiments and summarize our main results.

Because the efficiency gains of OMD are surprisingly small when the moments are homoscedastic (but correlated) unless there are restrictions across the parameters of the model, the interesting model for OMD involves correlated moments in which the structure of the correlation is known and aids in the identification of the parameter. We generate $D_p = (Z_p + \rho Z_{p+1}) / (1 + \rho)^{.5}$ for $p = 1$ to 10 from a set of mean 0, variance θ , iid random variables $Z_1, \dots, Z_{11}$. The D variables are mean 0 with variance θ , but D_p and D_{p+1} have a covariance of $\theta\rho/(1 + \rho^2)$. We assume that ρ is known, so the 10 sample variances and 9 first-order autocovariances are functions of the single unknown population variance θ , which is set to 1 in our experiments. The mix of variances and covariances in the vector of sample moments, m , implies heteroscedasticity in addition to serial correlation. AS reported simulation results from this model applied to a variety of distributions, sample sizes, and implied values for $\text{cov}(D_p, D_{p+1})$. (We were surprised that the theoretical efficiency gains of OMD over EWMD are relatively small for the preceding design even when ρ is 1.) We also investigated how increasing the size of the model while holding constant the number of moments that contain information on a given sample moment affects small-sample bias.

The results confirm our earlier finding that the bias in OMD is larger when the sample size is small or the distribution is poorly behaved. For example, in the lognormal case when the sample size is 100 and ρ is .5, the bias is $-.551$, and the bias is $-.074$ in the corresponding case for the normal. Second, EWMD continues to dominate OMD in RMSE and MAE. Third, the small-sample bias rises with the size of the model while holding constant the number of moments that contain information on a given sample moment.

4.4 The Performance of IWOMD

We have already noted that the experiments with independent, homoscedastic moments in Table 1 confirm that IWOMD is unbiased. We now provide a broader assessment of the performance of this estimator. Columns (10)–(14) of Table 1 report the bias, RMSE and std (they are the same for unbiased estimators), MAE, and 90% coverage rate for IWOMD for the homoscedastic model. The MAE

and RMSE of IWOMD are less than or equal to that of OMD in 33 of 36 cases and 30 of 36 cases, respectively. For the $t(5)$ distribution with 50 observations, the MAE values are .201 for OMD and .125 for IWOMD. The RMSE values are .223 and .202, respectively. On the other hand, the RMSE of the IWOMD estimator is always greater than EWMD.

Comparison of the RMSE and MAE between EWMD and IWOMD provides evidence of the cost of estimating the weighting matrix that is uncontaminated by the correlation between the weights and the second moments. For poorly behaved distributions and small sample sizes, the costs can be quite large. For example, for the $t(5)$ distribution with $N = 50$ the MAE for EWMD is .071 and .125 for IWOMD. The difference corresponds to an increase of more than 75% and the RMSE rises from .126 to .202. The difference dissipates as sample size increases but there is still a 50% difference in RMSE for the same experiment using 500 observations to estimate each moment.

AS reported Monte Carlo experiments for IWOMD based on 10 moments from two different distributions. The detailed results showed that IWOMD never outperforms EWMD in RMSE or MAE when N is 50 or 100. As the sample size increases the two estimators perform similarly, with IWOMD having an advantage in larger sample sizes for pairs of distributions that imply a substantial asymptotic efficiency gain for IWOMD and OMD. The sampling variance of IWOMD is larger than OMD even though the asymptotic variances of IWOMD and OMD are identical. As in the single-distribution case, the cost of estimating the weighting matrix is high. The experiments of AS based on serially correlated data show that IWOMD often dominates OMD in MAE and RMSE but usually does not perform as well as EWMD unless the efficiency gain is above 1.4 and the sample size is above 300.

In columns 13 and 14 of Table 1 we examine statistical inference based on IWOMD. For each replication we computed the 90% confidence-interval estimate using $\hat{\Omega}$ to evaluate the asymptotic standard-error formula (which is the same as OMD). The coverage rate for 90% confidence intervals based on IWOMD [col. (13)] are typically closer to .9 than OMD but well below the coverage rates for EWMD. This may seem puzzling given the very good performance of the EWMD confidence intervals based on the asymptotic theory. The source of the problem is that the std of IWOMD is often much larger than the asymptotic standard error (not reported). OMD and IWOMD are designed to reweight the sample moments to correct for heteroscedasticity and serial correlation that is reflected in Ω . The IWOMD asymptotic standard errors ignore the fact that sampling error in $\hat{\Omega}_g$ introduces additional heteroscedasticity and serial correlation into the weighted sample moments. (In the OMD case the correlation between the sampling error in $\hat{\Omega}$ and ε seems to reduce the discrepancy between the asymptotic standard errors and actual std.) The coverage rates for IWOMD are directly related to the ratio of the std of EWMD (true OMD for the experiments in Table 1) to the std of IWOMD and are close to .9 when this ratio is close to 1.

Table 2. Estimates of the Abowd-Card Stationary Covariance Structure of the Changes in Log Hours and Log Earnings for a PSID 11-Year Sample

Moment and estimator	Lag: $\tau = 0$	Lag: $\tau = 1$	Lag: $\tau = 2$
$\text{cov}(\Delta \text{earnings}_t, \Delta \text{earnings}_{t-\tau})$			
EWMD	.175	-.060	-.008
OMD	.086	-.026	-.008
IWOMD	.149	-.060	-.007
EWMD (Abowd and Card)	.172	-.060	-.007
$\text{cov}(\Delta \text{hours}_t, \Delta \text{hours}_{t-\tau})$			
EWMD	.131	-.047	-.006
OMD	.060	-.022	-.005
IWOMD	.111	-.043	-.005
EWMD (Abowd and Card)	.117	-.035	-.011
$\text{cov}(\Delta \text{hours}_t, \Delta \text{earnings}_{t-\tau})$			
EWMD	.080	-.026	-.008
OMD	.026	-.002	-.005
IWOMD	.060	-.024	-.006
EWMD (Abowd and Card)	.073	-.023	-.006
$\text{cov}(\Delta \text{earnings}_t, \Delta \text{hours}_{t-\tau})$			
EWMD	.080	-.024	-.002
OMD	.026	-.005	-.003
IWOMD	.060	-.016	-.001
EWMD (Abowd and Card)	.073	-.020	-.002

For all Table 1 experiments, we computed bootstrap confidence intervals [col. (14)]. These performed much better but still have coverage rates that are usually between .75 and .85.

We have also examined the coverage rates of IWOMD confidence intervals for designs involving moments from two different distributions. We again find that coverage rates for the IWOMD bootstrap confidence intervals are better than OMD but not as good as EWMD (see AS for details). We also find that the bootstrap confidence intervals perform better than confidence intervals constructed from the conventional asymptotic standard errors, which are too small. (Results are not reported.)

4.5 Summary

The experiments based on independent and homoscedastic second moments in Table 1, moments from two different distributions, and correlated moments all indicate, first, that θ_{OMD} suffers from serious small-sample bias in many cases; second, that it is almost always dominated by θ_{EWMD} in RMSE, MAE, and the coverage rates for 90% confidence intervals; and, third, that there is a large cost to having to estimate the weighting matrix. θ_{EWMD} also typically dominates θ_{IWOMD} , which is unbiased but has a larger sampling variance.

5. AN EMPIRICAL EXAMPLE

It is difficult to know what to assume about the data and models used in practice. Therefore, we supplement the preceding Monte Carlo analysis by applying the various estimators to Abowd and Card's (1987, 1989) analysis of the covariance of changes in log earnings and changes in log hours from the Panel Study of Income Dynamics (PSID). The data are based on an 11-year (1969–1979) sample of male heads of household. The sample consists of 1,536 individuals with complete data for the period, annual hours

continuously above 0 but less than or equal to 4,680, and average hourly wages continually less than \$100. [In constructing our sample, we followed the data appendix of Abowd and Card (1987), who worked with a sample size of 1,448. With their assistance we attempted to track down the differences between their sample and ours. We gave up after a modest effort because the EWMD and OMD estimates are very similar for the two samples.] All dollar values have been adjusted to 1967 levels using the Consumer Price Index. Log hours and log earnings are adjusted for labor-market experience and year effects prior to computing variances. The adjustment consists of separate regressions of the change in log earnings and the change in log hours on the 1967 potential experience level (age – education attainment – 5) and time dummies. The remainder of the procedure is performed using the mean 0 residuals from the experience regression.

The assumption of stationarity in first differences implies that the autocovariances and cross-covariances do not depend on time. The raw data for 1969–1979 provide 10 observations per person on the changes in log earnings and in log hours. There are 210 unrestricted moment estimates covering the various years and lags (10 hours variances, 45 earnings autocovariances, 10 earnings variances, 45 autocovariances, and 100 hours/earnings covariances). The stationary model has 39 unique parameters (1 hours variance, 9 hours autocovariances, 1 earnings variance, 9 earnings autocovariances, and 19 hours/earnings covariances). Table 2 presents the EWMD and OMD estimates of the covariance at time lags of 0, 1, and 2. The results of Abowd and Card (1987) are presented as well.

The difference between EWMD and OMD is striking. There is a systematic tendency for the absolute value of the EWMD estimates to exceed that of the OMD estimates. For example, the OMD estimate of the variance of the change in log earnings is .086, which is less than half of the EWMD estimate. The OMD estimate of the covariance of the change in hours and the first lag of the change in log earnings is .026, which is less than a third of the EWMD estimate of .080. Altonji et al. (1987) reported the same phenomena for similar data on earnings and hours as well as for data on changes in family income, wage rates, consumption, and the hours of unemployment. The fact that the OMD estimates are consistently below the EWMD estimates in absolute value is disturbing but expected in view of the discussion in Section 2.

Note, however, that the IWOMD estimates are somewhat smaller in absolute value than EWMD despite the fact that both estimators are unbiased. This may indicate that the covariances are in fact nonstationary because in the nonstationary case the true weights are likely to be positively related to the absolute values of the variances and covariances. As a result, OMD and IWOMD produce weighted averages of the true variances and covariances that will be less than the simple average produced by EWMD.

We would like to compare the three estimators using a model of the Abowd-Card data that we know is true by construction. To this end, we perform an experiment in which we treat the Abowd-Card sample as a popula-

Table 3. Simulation of a True Covariance Model of the Change in Log Hours and Earnings Based on 11-Year PSID Sample

Moment	EWMD				OMD					IWOMD		
	Std RMSE	MAE	90% coverage		Bias	Std	RMSE	MAE	90% asym. coverage	Std RMSE	MAE	90% asym. coverage
			Asym.	Boot								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
var($\Delta \text{earnings}_t$)	.108	.075	.86	.89	-.722	.044	.723	.722	.00	0.258	.171	.08
cov($\Delta \text{earnings}_t, \Delta \text{earnings}_{t-1}$)	.156	.104	.87	.88	-.737	.051	.739	.739	.00	0.314	.215	.08
cov($\Delta \text{earnings}_t, \Delta \text{earnings}_{t-2}$)	.511	.366	.90	.89	-.808	.162	.824	.812	.00	1.140	.761	.08
var(Δhours_t)	.114	.077	.88	.89	-.728	.045	.730	.729	.00	0.256	.150	.07
cov($\Delta \text{hours}_t, \Delta \text{hours}_{t-1}$)	.159	.108	.87	.87	-.725	.049	.726	.727	.00	0.301	.180	.09
cov($\Delta \text{hours}_t, \Delta \text{hours}_{t-2}$)	.412	.269	.90	.90	-.778	.110	.786	.779	.00	0.766	.484	.13
var(Δhours_t)	.161	.110	.87	.87	-.804	.048	.806	.807	.00	0.309	.202	.09
cov($\Delta \text{hours}_t, \Delta \text{earnings}_{t-1}$)	.246	.170	.88	.88	-.800	.059	.802	.801	.00	0.392	.250	.08
cov($\Delta \text{hours}_t, \Delta \text{earnings}_{t-2}$)	.453	.319	.88	.88	-.851	.095	.856	.854	.00	0.718	.475	.10
var($\Delta \text{earnings}_t, \Delta \text{hours}_{t-1}$)	.279	.185	.87	.87	-.815	.061	.817	.819	.00	0.402	.284	.07
cov($\Delta \text{earnings}_t, \Delta \text{hours}_{t-2}$)	.624	.407	.86	.86	-.846	.101	.852	.854	.00	0.716	.476	.09

NOTE: 1. Estimates are based on 500 replications. 2. Columns (3), (9), and (12) report the coverage rate of a symmetric asymptotic confidence interval. Columns (4) and (13) report the coverage rate of a 90% symmetric bootstrap confidence interval. See Table 1, note 2.

tion and resample from it with replacement. We specify the following “model” of the sample moments. Consider first the variance in hours. Let $m_{t,t-\tau}$ be the estimate of $\text{cov}(\Delta \text{hours}_t, \Delta \text{hours}_{t-\tau})$ for a sample drawn from the population with replacement. Then

$$m_{t,t-\tau} = X_{t,t-\tau} \theta_\tau + \varepsilon_{t,t-\tau},$$

$$t = 1970, \dots, 1979, \tau = 0, \dots, 9, \quad (12)$$

where $X_{T,t-\tau}$ is $\text{cov}(\Delta \text{hours}_t, \Delta \text{hours}_{t-\tau})$ for the “population.” The true parameter value for θ_τ is 1, and the sampling error $\varepsilon_{t,t-\tau}$ has mean 0. We construct a corresponding model for the variance of earnings and the covariance of hours and earnings (leads and lags). The observations are stacked into the model

$$m = X\theta + \varepsilon, \quad (13)$$

where m is (210×1) , X is a (210×39) matrix of constructed “explanatory variables,” θ is a (39×1) vector of parameters, and ε is a (210×1) vector of sampling errors. For example, if we order the observations so that the first 10 correspond to the observations on $\text{cov}(\Delta \text{hours}_t, \Delta \text{hours}_{t-\tau})$, then the first 10 elements of m are the “sample” estimates of the “population” parameters $\text{cov}(\Delta \text{hours}_{1970}, \Delta \text{hours}_{1970}) \dots \text{cov}(\Delta \text{hours}_{1979}, \Delta \text{hours}_{1979})$. The first 10 rows of the first column of X are $\text{cov}(\Delta \text{hours}_{1970}, \Delta \text{hours}_{1970}) \dots \text{cov}(\Delta \text{hours}_{1979}, \Delta \text{hours}_{1979})$ for the “population.” All of the elements of the true θ equal 1. The dimensions of the constructed model match the Abowd and Card stationary model.

We draw samples of 500 observations (with replacement) from the population of 1,536 individuals and estimate the 39 elements of θ by EWMD, OMD, and IWOMD. In Table 3 we report the bias, std, RMSE, MAE, and coverage rates for 90% confidence intervals for a subset of the moments. We focus our attention on the results for autocovariances and cross-covariances at lags 0, 1, and 2. The results are quite striking. OMD is badly biased and has a much larger RMSE and MAE than EWMD. The bias is typically be-

tween $-.7$ and $-.85$, which is very large relative to the true parameter value of 1. These results suggest that the main reason that the OMD estimates of the “stationary model” of hours and earnings in Table 2 are smaller than the EWMD and IWOMD estimates is bias, not misspecification.

The RMSE of the OMD estimator typically exceeds the RMSE for EWMD by a factor between 2 and 7. The MAE is also much larger for OMD. OMD does usually have a smaller sampling variance, but its advantage in this dimension is not enough to make up for the large bias. Using the diagonal of $\hat{\Omega}$, instead of the complete $\hat{\Omega}$, in the stationary model in Table 2 and the constructed model of Table 3 leads to a substantial reduction in the bias (not reported). We speculate that OMD performs particularly poorly when Ω is a large matrix and the random variables are correlated, which is consistent with the findings mentioned in Section 4.3.

In column (3) of Table 3, we report the probability that the 90% EWMD confidence-interval estimate based on the asymptotic formulas (using $\hat{\Omega}$) contains the true value of 1. In the case of $\text{var}(\Delta \text{earnings}_t)$, the coverage rate is .86. The coverage rate is typically about .88 for other moment parameters. Column (4) of Table 3 reports on the coverage rate based on a 90% symmetric bootstrap confidence interval. In the case of $\text{var}(\Delta \text{earnings}_t)$, the coverage rate is .89, and the typical result is that bootstrap-based confidence intervals and conventional asymptotic confidence-interval estimates perform about the same and cover the true value of 1 slightly less often than .90. Whether or not the departure of the actual size from the true size is serious would seem to depend on the application.

In contrast, we reject the hypothesis that the OMD estimates equal the true value in every single replication of the experiment [col. (9)], which is a reflection of the fact that the bias in OMD is many std's away from 0. [Compare cols. (5) and (6).] The coverage rates for IWOMD, computed using the same asymptotic approximation as in OMD, are also very poor because the asymptotic approximation understates the sampling variance of IWOMD. We suspect that bootstrap IWOMD confidence intervals would

perform better than the asymptotic confidence intervals but did not compute them because of the computational burden.

In summary, EWMD is the best estimator for fitting linear models of the covariance structure of the PSID data on earnings and hours. The relative performance of EWMD, OMD, and IWOMD in the PSID data is consistent with the simulations in Table 1, but the problem of bias in OMD seems far worse in real-world data with a large model. The results provide at least some evidence that the small-sample bias in OMD is a serious problem in applications.

The results of Schwert and Seguin (1990) provide another illustration of the concerns we raise. They analyzed the covariance in stock-market returns in a microeconomic model relating the second moments of stock portfolio returns to a constant and a time-varying measure of aggregate market volatility. They estimated the model individually for each second moment from five size-ranked portfolios (5 variances and 10 covariances) by OLS. Then the OLS residuals were used in a Glejser-style regression relating the square of the residuals to the regressors. Feasible weighted least squares (WLS) is applied using the predicted values from the second-stage regression. Ignoring the aggregate-market-volatility regressor, the Glejser procedure is equivalent to estimating the variance of the sample second moments using the average squared deviation of the second moments from their average. Twenty-nine of the thirty WLS coefficients are smaller in absolute value than the OLS coefficients even though both procedures yield consistent estimates. This is what our analysis predicts, although the difference between the WLS and OLS results is not sufficient to change their conclusions.

6. CONCLUSION

In this article we provide a theoretical argument and Monte Carlo evidence showing that OMD is biased in small samples. For a given sample size, the bias depends on the distribution of the underlying data. The bias is worse when the data are drawn from distributions with heavy tails. The problem goes away as the sample size gets large but does not go away (and may get worse) as the number of moments available to fit a model increases or as the size of the model increases, holding constant the number of moments that are informative about a given parameter. Our findings using the Abowd-Card data are particularly striking. We also present an estimator called IWOMD, which is an unbiased split-sample alternative to conventional OMD and is asymptotically equivalent to it. In most cases we consider, however, the asymptotic efficiency gain of IWOMD relative to EWMD is overwhelmed by the extra noise introduced by estimated weights.

Our conclusion is that EWMD is almost always preferable to using OMD when the optimal weighting matrix is unknown and unconstrained, especially when bias is an important concern. This is true even in situations in which OMD is far superior in asymptotic efficiency. At a minimum, researchers should estimate models by both OMD and EWMD, or both OMD and IWOMD, and worry about bias in OMD if the parameter estimates differ substan-

tially. Discrepancies between OMD and EWMD may be a risky basis for Hausman-type model-misspecification tests for models of second moments.

If one is going to use EWMD because of small-sample considerations, the issue of statistical inference arises. The limited Monte Carlo evidence in Table 1 and some additional evidence for experiments involving moments from two different distributions of AS suggests, perhaps surprisingly given our evidence on bias in OMD, that in most cases the standard asymptotic formula provides a satisfactory basis for inference when using EWMD. Hypothesis testing in the case of OMD is a disaster in small samples with poorly behaved distributions because the sampling distribution of OMD is shifted far away from the true parameter value. IWOMD confidence intervals also have lower coverage rates than EWMD.

We wish to highlight several areas for future research. First, perhaps the use of robust estimation methods to estimate the weighting matrix or to estimate the moments being modeled (see Koenker et al. 1994) may lead to an OMD estimator that is superior to conventional OMD and IWOMD. Second, it is clear from the information in Figure 2 and the more detailed results of AS about the performance of true OMD that prior information about the appropriate weighting matrix is valuable. Iterative or continuous updating of the weighting matrix that takes advantage of the link between the weighting matrix and the moments being fitted (see Hanson, Heaton, and Yaron 1996) is one possibility in this vein. These techniques use the moment estimates to improve the estimates of the weighting matrix. The techniques do not, however, necessarily eliminate the covariance between the weights and the moment estimates. There may be many situations in which researchers may improve on EWMD and OMD by using prior information about which sets of moments are likely to be highly correlated or particularly noisy to reduce the dimensionality of $\hat{\Omega}$.

Third, it may be worth investigating estimators that converge to EWMD as the number of moment conditions P becomes large with N fixed and to IWOMD and OMD as N becomes large with P fixed. For example, in a model of uncorrelated heteroscedastic moments we might consider an estimator formed from a weighted sum of the IWOMD and EWMD estimates. The respective weights are the adjusted R^2 and one minus the adjusted R^2 from a first-stage regression of the diagonal elements of $\hat{\Omega}^{-1}$ estimated from one-half of the sample on the corresponding elements from the other half of the sample. The adjusted R^2 is near 0 in small samples, placing the majority of the weight on EWMD. As the sample size increases, the adjusted R^2 tends toward 1, increasing the weight on IWOMD. Initial simulation results suggest that the weighted estimator often has better properties than EWMD and IWOMD.

Fourth, our Monte Carlo evidence is focused on cases in which the model of the second moments is linear in parameters. Many applications of OMD involve models in which the parameters of interest are nonlinear functions of the second moments. (For example, Abowd and Card estimated factor models that are nonlinear in the second moments.) In these cases, bias is likely, but the sign and severity is not

likely to be easy to predict. For example, if a factor loading parameter is estimated as the covariance divided by a variance, then downward bias (in absolute value) in the two moments may partially cancel out, and the direction and size of the parameter bias will depend on the relative bias in the two moments. Clark (1996) extended our analysis to consider three nonlinear models that are popular in the literature and found serious problems with OMD. Further work on nonlinear models is needed.

Fifth, Monte Carlo evidence is needed on the validity of model misspecification tests based on OMD and EWMD, an issue that we have not explored. Sixth, we suspect similar biases in maximum likelihood and quasi-maximum likelihood estimation of covariance structures. This should be investigated.

We wish to emphasize that our basic concern—that feasible GMM estimators are biased in small samples because of correlation between the moments used to fit the model and the weight matrix—applies in situations involving moments of any order. This is not a new point because it is well known, for example, that feasible GLS is biased in small samples in many situations. It has not gotten the attention it deserves, however, in the burgeoning literature that involves GMM. We suspect that the problem of bias is far more serious in models of covariance structures than in models of conditional mean functions (such as the standard regression model with or without endogenous regressors) that are more common in applied work. In such models the errors are often close to symmetric and the weighting matrices involve second moments rather than fourth moments. Whether GMM is in fact better in RMSE than estimators based on simpler weighting schemes seems to us to be an open question even in these applications. In any case, a careful investigation of the small-sample properties of GMM in a variety of contexts would seem to be warranted.

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APPENDIX: THE DIRECTION OF THE BIAS

This appendix establishes that $\text{cov}(\varepsilon_p, \omega_p)$ is negative provided that the distribution of D_{ip}^2 is skewed to the right, with $E[(D_{ip}^2 - E(D_{ip}^2))^3] > 0$. Assume that the underlying data, D_{ip} , are known to have mean 0. Let μ_{jp} represent the j th population moment of D_{ip} , and let m_{jp} be the j th sample

moment of D_{ip} , computed as

$$m_{jp} = \frac{1}{N} \sum_{i=1}^N D_{ip}^j. \quad (\text{A.1})$$

To simplify the notation we suppress the p subscripts in what follows so that m_1 is a sample mean and m_2 is a sample variance. The notation m_j should not be confused with m_p used in the text to represent the p th element of the vector m of second moments. Note that

$$E(m_j) = \mu_j \quad (\text{A.2})$$

and

$$m_2 = \mu_2 + \varepsilon, \quad (\text{A.3})$$

where ε is the sampling error. The variance of m_j is

$$\text{var}(m_j) = \frac{1}{N} (\mu_{2j} - \mu_j^2). \quad (\text{A.4})$$

The variance of m_2 is

$$\text{var}(m_2) = \frac{1}{N} (\mu_4 - \mu_2^2), \quad (\text{A.5})$$

which is estimated as

$$\omega = \frac{1}{N} (m_4 - m_2^2). \quad (\text{A.6})$$

The covariance between ε and ω depends on the skewness of D^2 as shown by

$$\begin{aligned} \text{cov}(\varepsilon, \omega) &= E(\varepsilon\omega) - E(\varepsilon)E(\omega) \\ &= E(\varepsilon\omega) \\ &= E(m_2\omega) - E(\mu_2\omega) \\ &= \frac{1}{N} E(m_2m_4 - m_2^3) - \frac{(N-1)}{N^2} (\mu_2\mu_4 - \mu_2^3) \\ &= \frac{(N-1)}{N^3} (\mu_6 - 3\mu_2\mu_4 + 2\mu_2^3) \\ &= \frac{(N-1)}{N^3} E[(D^2 - \mu_2)^3], \end{aligned} \quad (\text{A.7})$$

where the last equality follows from the fact that

$$\begin{aligned} \mu_6 - 3\mu_2\mu_4 + 2\mu_2^3 &= E[(D^4 - 2D^2\mu_2)(D^2 - \mu_2)] \\ &= E[(D^4 - 2D^2\mu_2 + \mu_2^2)(D^2 - \mu_2)] \\ &\quad - E[\mu_2^2(D^2 - \mu_2)] \\ &= E[(D^2 - \mu_2)^3]. \end{aligned} \quad (\text{A.8})$$

Thus, the expectation of $\text{cov}(\varepsilon, \omega)$ is positive if the distribution of D^2 is skewed to the right. This restriction is satisfied for all of the distributions we consider. It is not satisfied if the density of D is symmetric and the density is concentrated near the minimum and maximum of D . For example, consider the density $dF(D) = c|D|^\alpha$ when $-D_b \leq 0 \leq D_b$ and 0 otherwise, where c normalizes the cdf to 1 and $\alpha > 0$. When $\alpha > 0$, the density has a minimum at 0 and maximums at $-D_b$ and D_b . We established numerically that the

sign of $E[(D^2 - \mu_2)^3]$ is less than 0 when $\alpha > 1$ and that there is a small positive bias in θ_{OMD} when this distribution is used in an experiment analogous to those in Table 1. The intuition is that, when α is large, the "unusual" observations on D are those near the mean of 0. These correspond to low values of D^2 but substantial values of $(D^2 - \mu_2)^2$, which is the contribution to the weighting matrix. Consequently, second moments with negative sampling errors receive too little weight, and θ_{OMD} is biased upward.

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